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Modal Analysis of a Cantilever Beam

ADVANCED DYNAMICS OF MECHANICAL SYSTEMS
LAB ACTIVITY REPORT

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1 | Data of the reference structure

1.1. geometry and material properties

The given structure is a cantilever beam, with the left end clamped and the right end that is free to move in the vertical plane. As shown in the Figure 1.1, the beam has a **rectangular cross section** and it's made of **aluminum**.

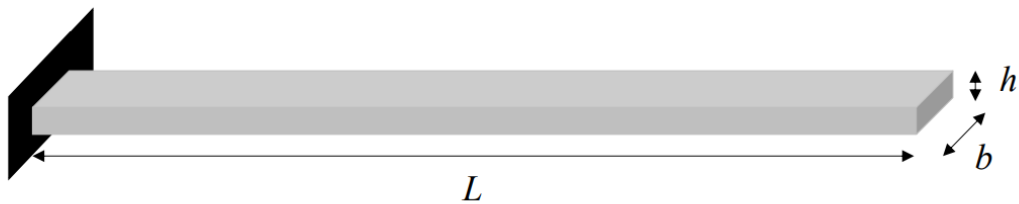


Figure 1.1: Beam schematic representation

Parameter	symbol	unit	value
Length	L	mm	1200
Thickness	h	mm	8
Width	b	mm	40
Density	ρ	kg/m^3	2700
Young's Modulus	E	GPa	68

Table 1.1: Beam's properties

Moreover, from the given data it's possible to compute the **moment of inertia**, J.

$$J = \frac{1}{12} \cdot b h^3 = 1706.77 mm^4$$

In order to proceed in the analysis, from now on all the parameters will be expressed in SI units.

2 | Vibration Modes

2.1. Standing wave solution

The system under analysis is composed by one single beam. The expression for the standing wave solution, that represent the transverse vibration in the vertical plane is known, and it's given by the following relation:

$$w(x, t) = [A \cos(\gamma x) + B \sin(\gamma x) + D \cosh(\gamma x) + B \sinh(\gamma x)] \cdot \cos(\omega t + \phi)$$

Where the wave number γ is linked to the frequency ω by the relation:

$$\gamma^4 = \frac{m}{EJ} \cdot \omega^2$$

The value of the constants A, B, C, D is now unknown, and will result as an output from the process of identification of the mode shapes.

2.2. Boundary conditions

In order to define the four missing constants we need 4 boundary conditions.

CLAMPED END:

$w(0, t) = 0$ no displacement;

$\frac{\partial w}{\partial x}|_0 = 0$ no rotation;

FREE END:

$EJ \cdot \frac{\partial^2 w}{\partial x^2}|_L = 0$ no moment;

$EJ \cdot \frac{\partial^3 w}{\partial x^3}|_L = 0$ no shear;

2.3. Matrix formulation

Computing all the derivative it's possible to arrange these four equations in a **matrix** form, calling $[H(\omega)]$ the matrix of the coefficients, that is a function of γ (and so a function of ω). The unknowns A, B, C and D are stored in a column vector called $\underline{z}$. The system to solve is:

$$[H(\omega)] \cdot \underline{z} = \underline{0}$$

2.4. Computation of the natural frequencies

In order to solve the equation, we impose matrix $[H]$ to be singular. The operation:

$$\det[H(\omega)] = 0$$

results in a **characteristic equation** in the unknown ω . The equation is highly **non linear**: the solutions for the roots ω_i have to be obtained numerically.

Since the functions that are inside the characteristic equation are all periodic, infinite solutions are expected. We limit the research of the zeros of the characteristic equation to the first four ω_i , since we are interested in the range $[0; 200]$ Hz. In the following table are reported the values (in Hz and in Rad/s) for the first four omega founded:

ω	[Hz]	[Rad/s]
ω_1	4.50	28.30
ω_2	28.23	177.34
ω_3	79.03	496.57
ω_4	154.87	973.06

Table 2.1: Values for ω

It's is possible to have also a graphical feedback of the results, plotting in a semi-logarithmic scale the expression of the characteristic equation. The vertical lines on the plot shown in Figure 2.1 are in correspondence of the zeros founded numerically.

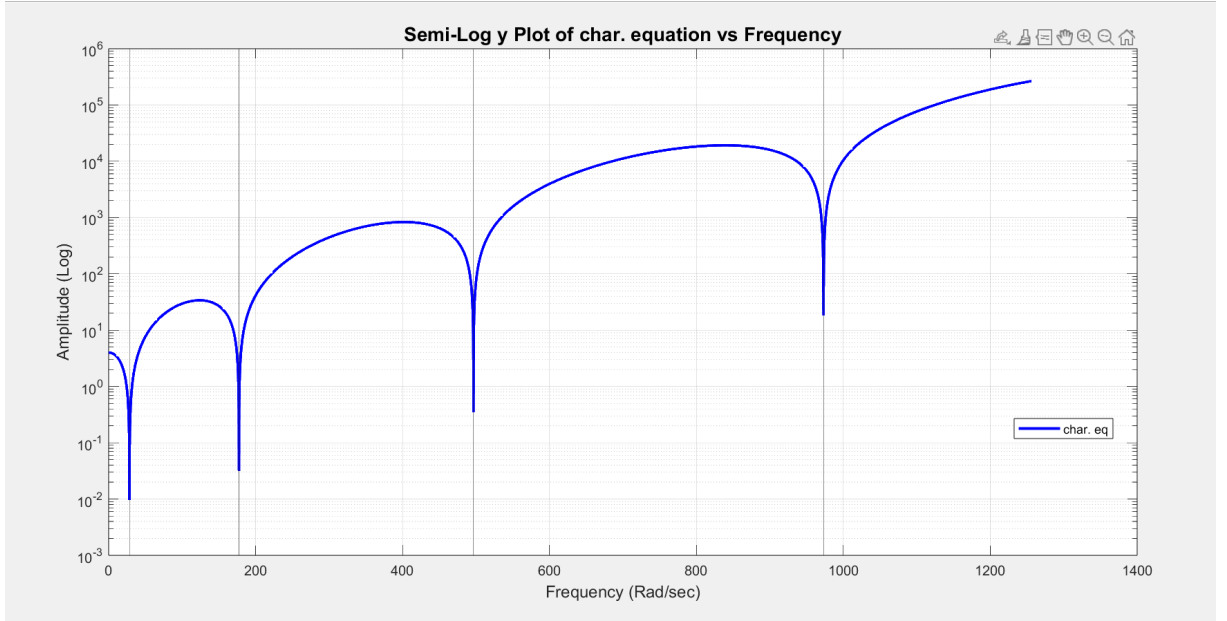


Figure 2.1: Plot showing the characteristic equation

2.5. Mode shapes computation and plot

For finding the vector $\underline{z}$ and define the **mode shapes** of the beam it is necessary to substitute back the ω_i founded into the system of equation:

$$[H(\omega_i)] \cdot \underline{z}^{(i)} = \underline{0}$$

The matrix $[H(\omega_i)]$ became now a constant matrix, and it's possible to evaluate the associate solution $\underline{z}^{(i)}$. Since we previously imposed the matrix $[H]$ to be singular, now we have that the equations in the system are **linearly dependent**: the solution for $\underline{z}^{(i)}$ is not unique, but defined unless a constant. We arbitrary decided to put $A^i = 1$ for every mode, and to compute the other coefficients normalized respect to A.

Solving the system for the four ω_i computed before we get four vectors $\underline{z}^{(i)}$. Substituting the coefficients $A^i = 1$, B^i , C^i , D^i in the standing wave solution, we can plot the shape of the first four modes, in the range $[0, 200]$ Hertz.

The plot of the modes are done considering a normalization on the y-axis, that can assume values bounded between $[-1, 1]$. From the plot, it's possible to note that the **clamped end** is correctly never subjected to displacement or rotation. The **number of nodes** is equal to the number i of the mode we are considering.

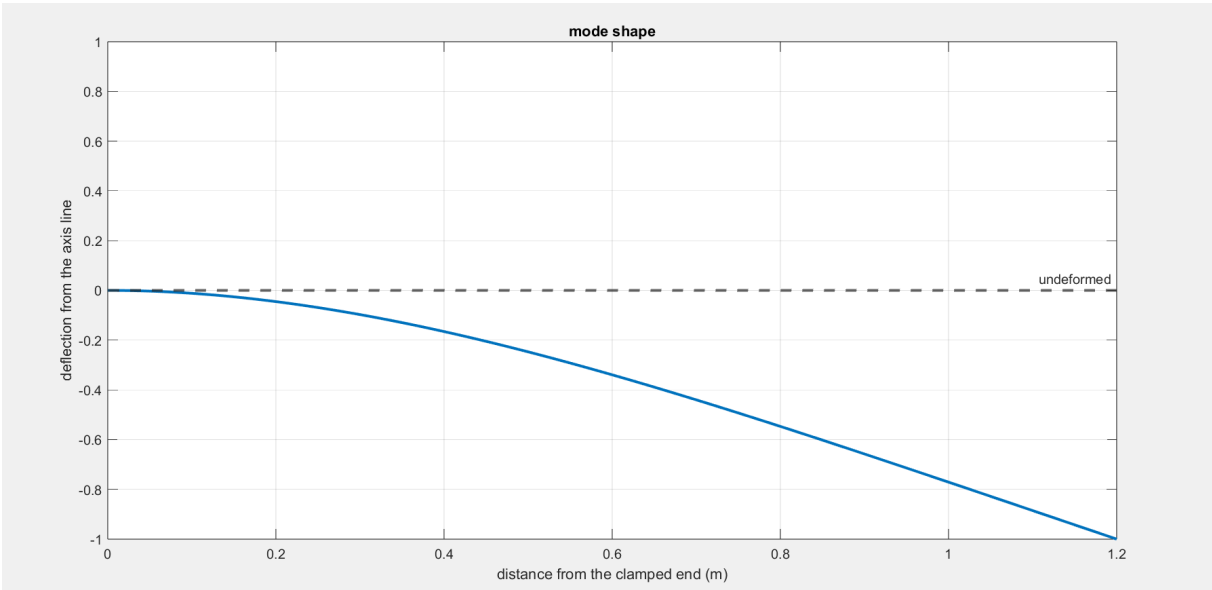


Figure 2.2: Mode 1

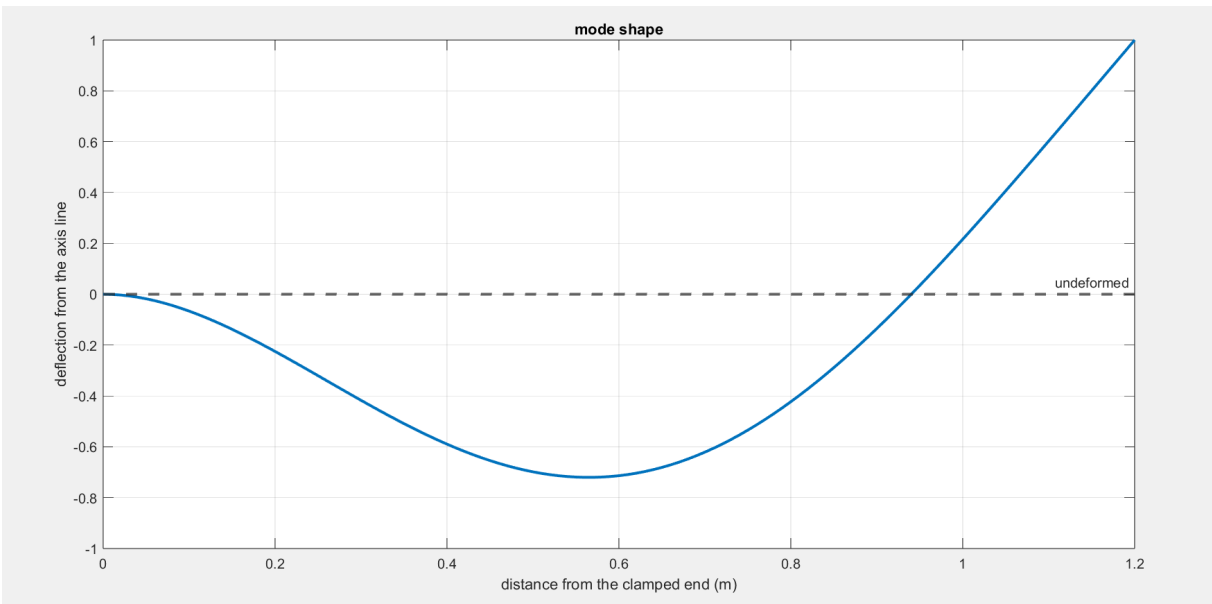


Figure 2.3: Mode 2

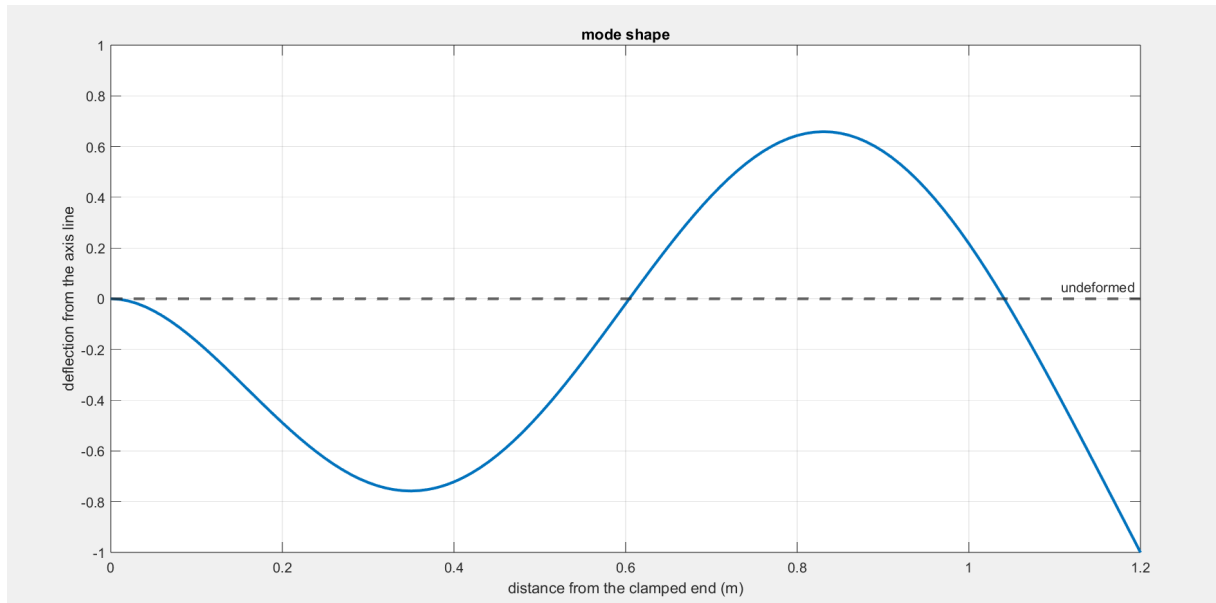


Figure 2.4: Mode 3

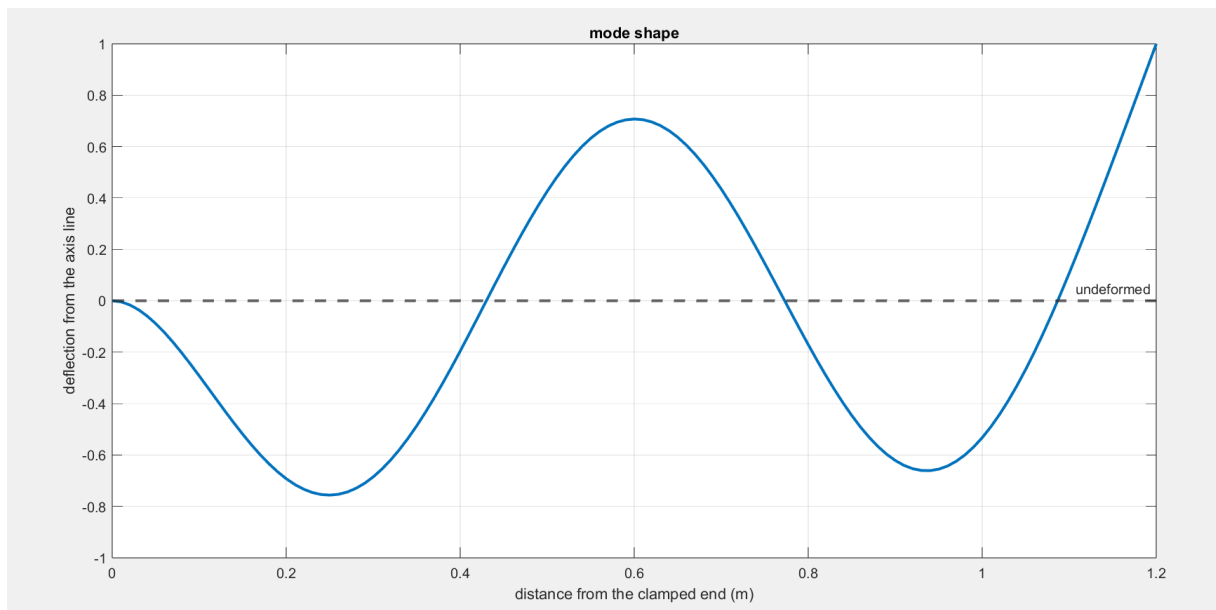


Figure 2.5: Mode 4

3 | Frequency Response Functions

3.1. General procedure

The Frequency Response Function (FRF) allows to relate the behavior of the beam (in terms of displacement of a specific point, x_j) to the action of an external force, applied in a generic x_k (inside the domain $[0, L]$). The force is supposed to be an **harmonic input**: $F_k(t) = F_0 \cdot e^{j\Omega t}$.

Relying on the modal approach, we can express the FRF for a generic configuration as it follows:

$$G_{jk}(\Omega) = \frac{x_{j0}}{F_{k0}} = \sum_{i=1}^4 \frac{\Phi(x_j)\Phi(x_k)/m_i}{-\Omega^2 + j2\xi_i\omega_i\Omega + \omega_i^2}$$

The modal mass m_i is computed as it follows:

$$m_i = \int_0^L m \Phi_i^2(x) dx$$

and it has been evaluated with the Matlab function *Trapz*.



Figure 3.1: Generic set-up analyzed

As a first example, it's reported the case in which the force is acting at a distance of $x_k = 1.2m$ from the clamped end and the sensor that reveals the displacement is placed is $x_j = 0.2m$. The FRF is plotted (Figure 3.2) both in magnitude and phase, since $G_{jk}(\Omega)$ is a **complex number**. In Figure 3.3 the single modes are visible: their sum provides

the overall FRF magnitude.

The **non-dimensional damping factor** ξ is supposed 0.01 (light damping).

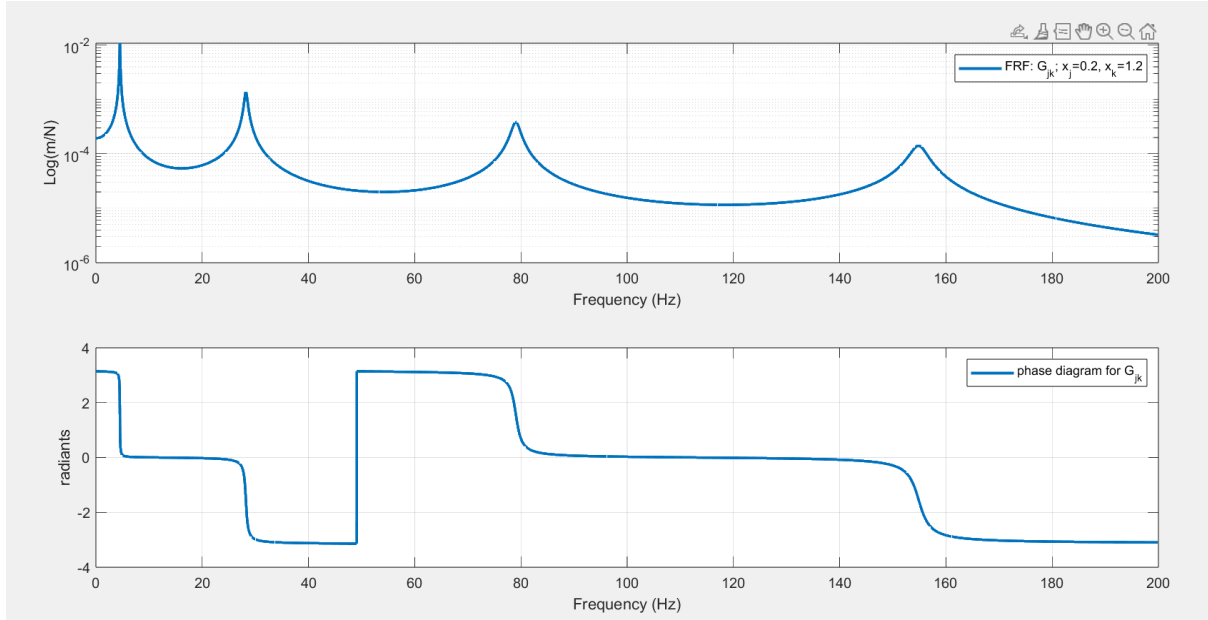


Figure 3.2: Frequency Response Function

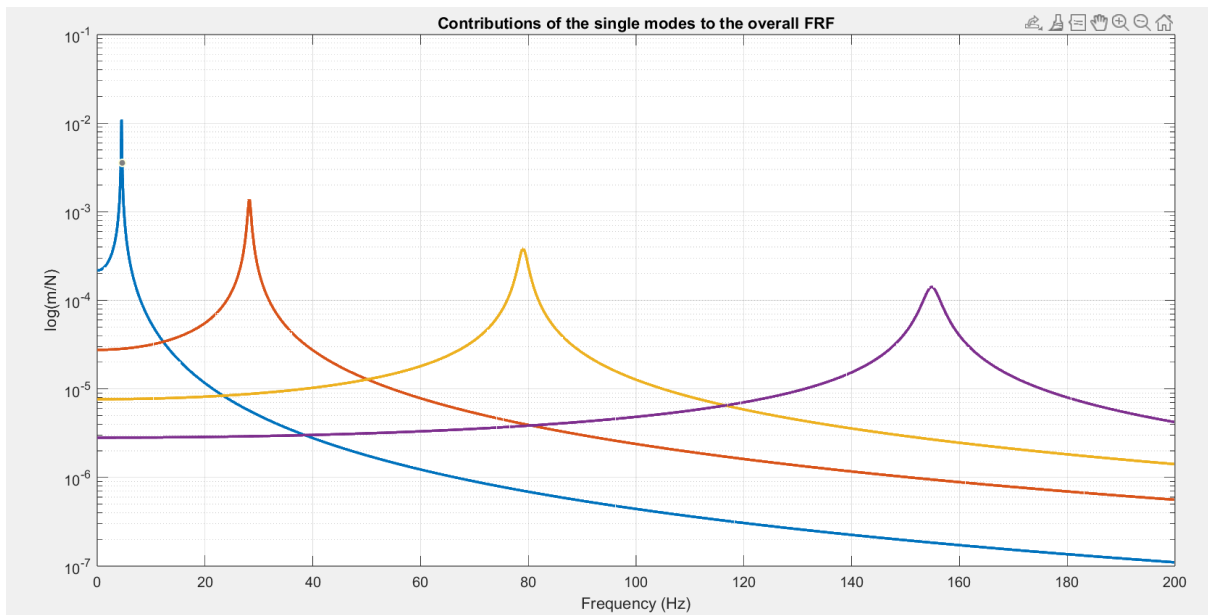


Figure 3.3: Contribution of the single mode i

3.2. Particular FRF analyzed

3.2.1. Co-Located FRF

A Frequency Response Function is said to be "**co-located**" when the output and the input point are the same. Mathematically:

$$x_j = x_k$$

From the theory, we expect to see an FRF with an alternation of poles and zeros. As an example, the position in which the FRF is evaluated (and the force applied) is assumed to be: $x = 0.2m$.

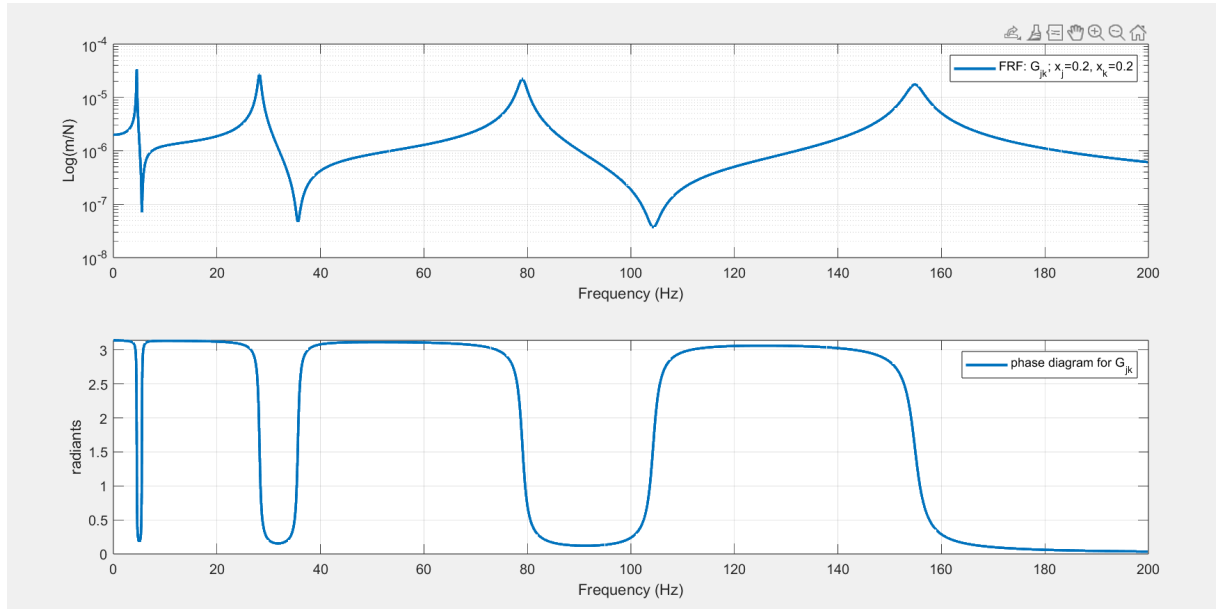


Figure 3.4: Co-located FRF ($x = 0.2m$)

3.2.2. Controllability problem

It is said that a mode is **non controllable** when the *force* is applied in a **vibration node** of that specific mode. The force is therefore not able to introduce energy in the system and to excite the specific mode. Applying the force in $x_k = 0.6043m$ and evaluating the response in $x_j = 0.2m$ we expect that the mode three will not appear in the graph.

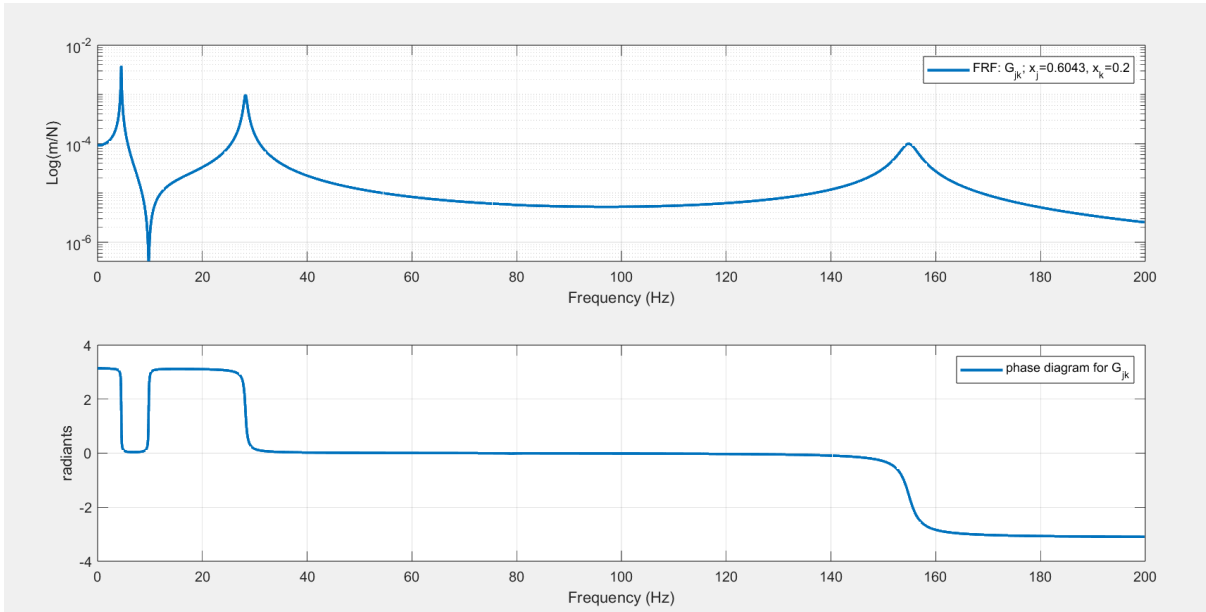


Figure 3.5: Controllability mode 3

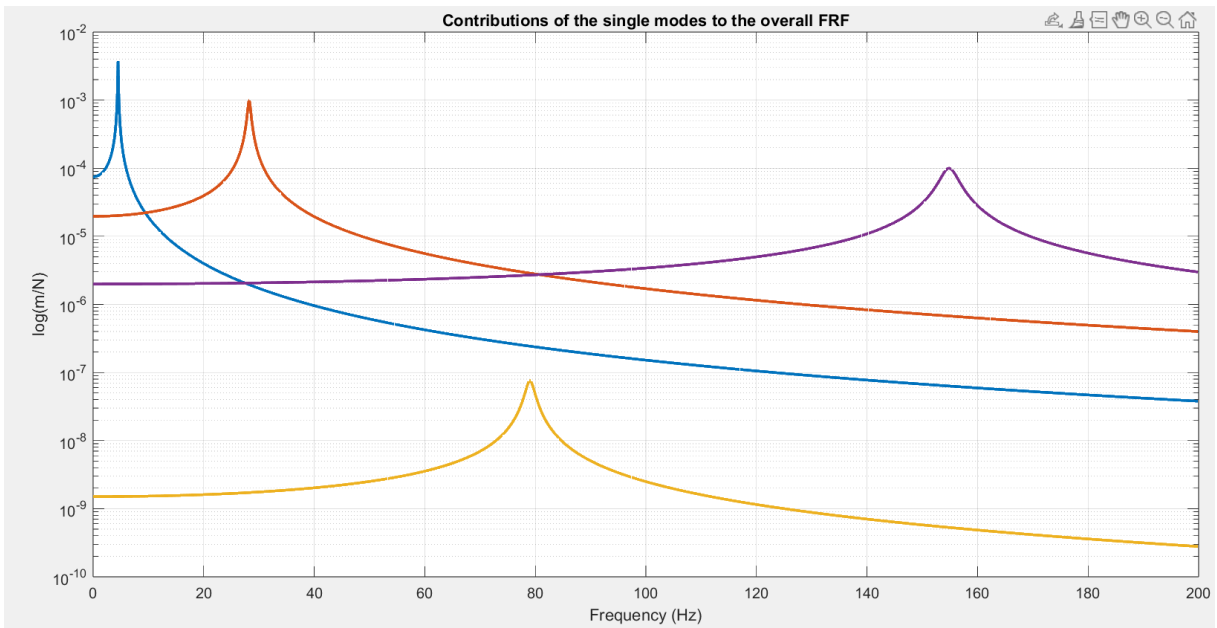


Figure 3.6: Controllability of mode 3: contributes

It is possible to perform a very similar analysis by putting the force in a point that is very close to the vibration node (like $x_k = L/2 = 0.6m$): we see that the resonance peak associated with mode number three is very small.

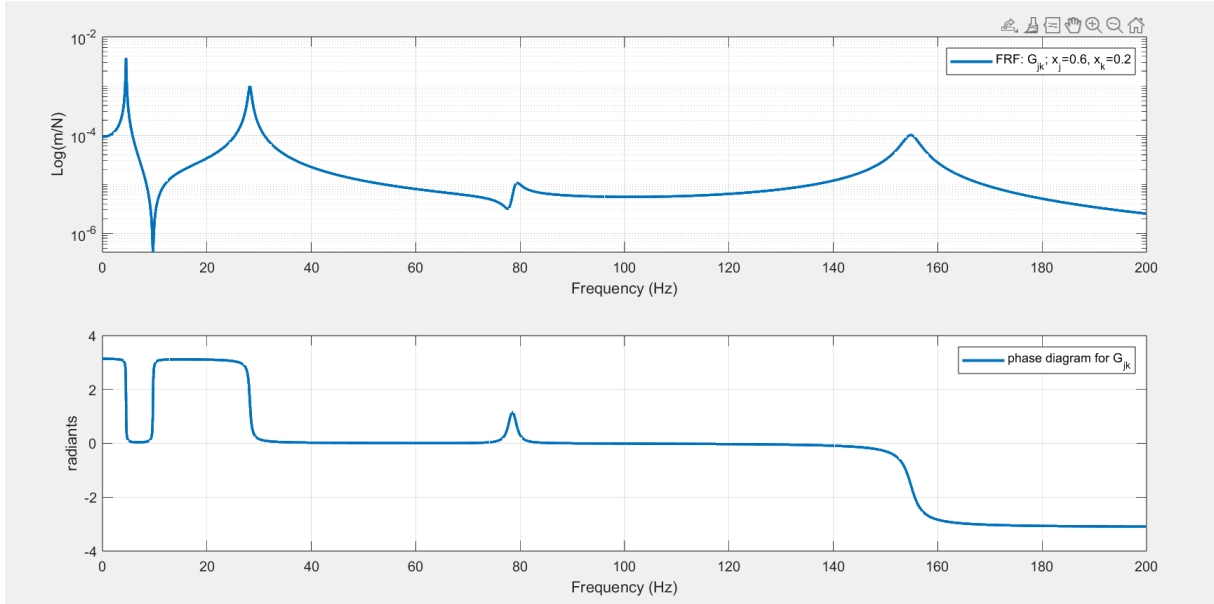


Figure 3.7: Controllability of mode 3

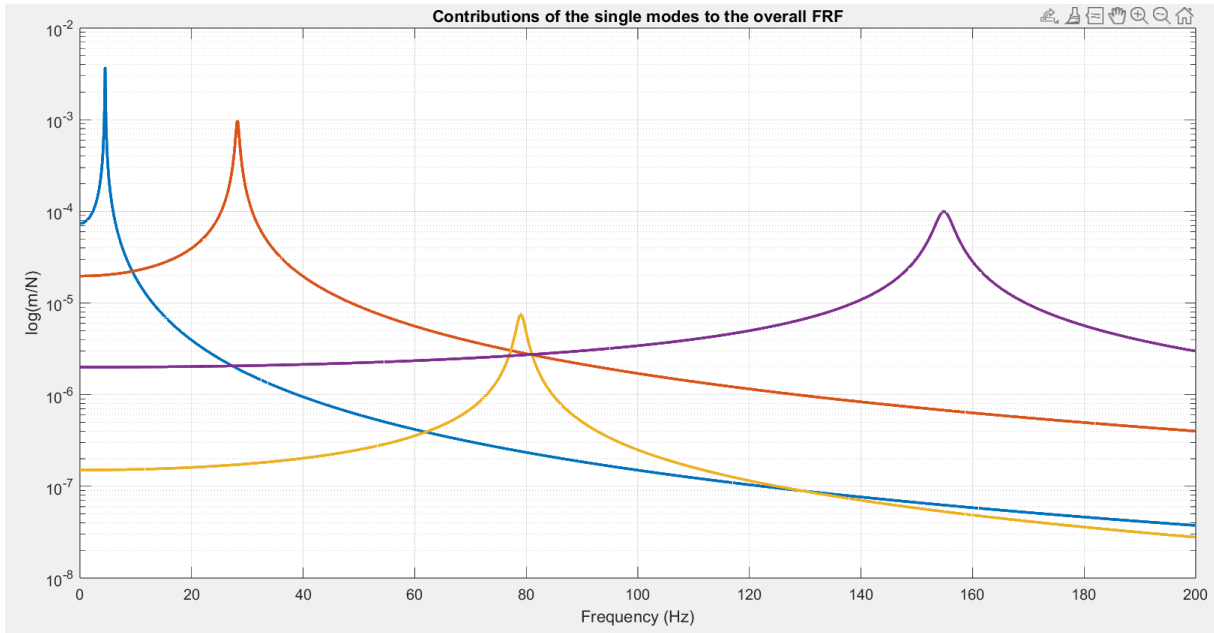


Figure 3.8: Controllability of mode 3: contributions

3.2.3. Observability problem

It is said that a mode is **not observable** when the *sensor* is applied in a **vibration node** of that specific mode. The force is therefore able to introduce energy in the system, but the position x_j of the looking point prevents us from seeing the contribution of that specific mode in the overall FRF. As an example, we put the force in $x_k = 0.55m$ (that is

an **anti-nodal point** for mode two) and the sensor in $x_j = 0.9401m$, a nodal point for mode two. We expect a plot (Figure 3.9) without the second resonant peak, even if mode two is actually excited by the force F_k .

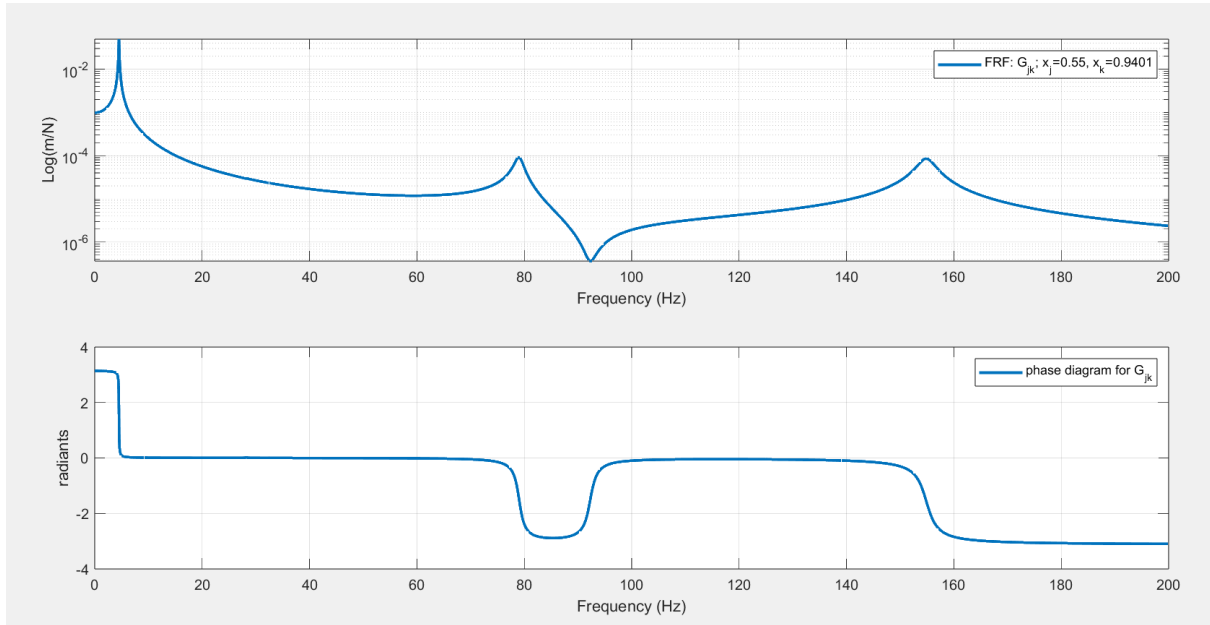


Figure 3.9: Observability of mode two

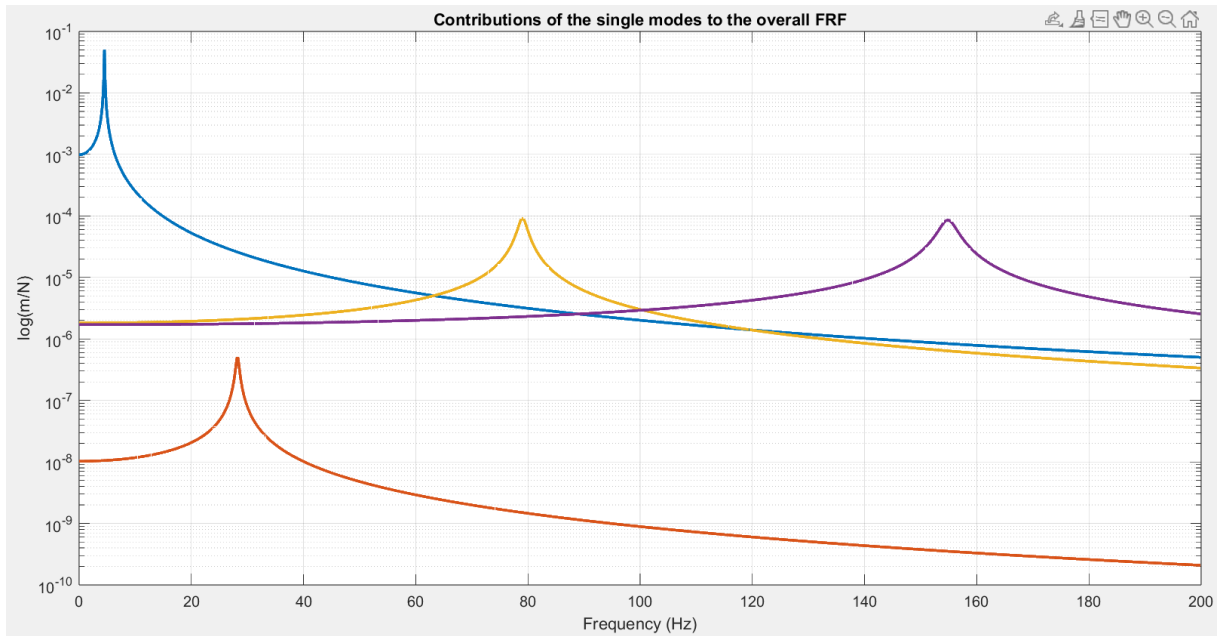


Figure 3.10: Observability of mode two: single contributes

3.2.4. Reciprocity principle

Reciprocity principle states that **swapping** the location of the sensor with the location in which the force is applied we get exactly the same Frequency Response Function. We verify this principle considering $x_j = 1.2m$ and $x_k = 0.2m$. What we expect to see in Figure 3.11 is a FRF that is the same of the FRF presented in Section 3.1, at the beginning of this Chapter (Figure 3.2).

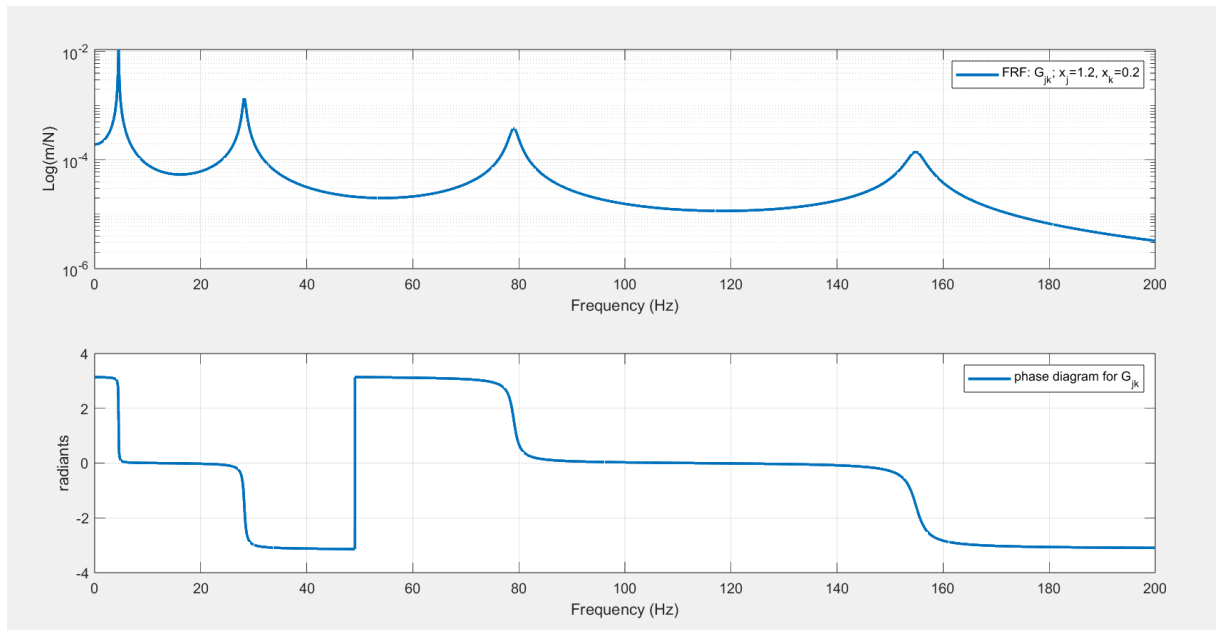


Figure 3.11: Reciprocity Principle

4 | Experimental identification

4.1. Definition of the Experimental Matrix

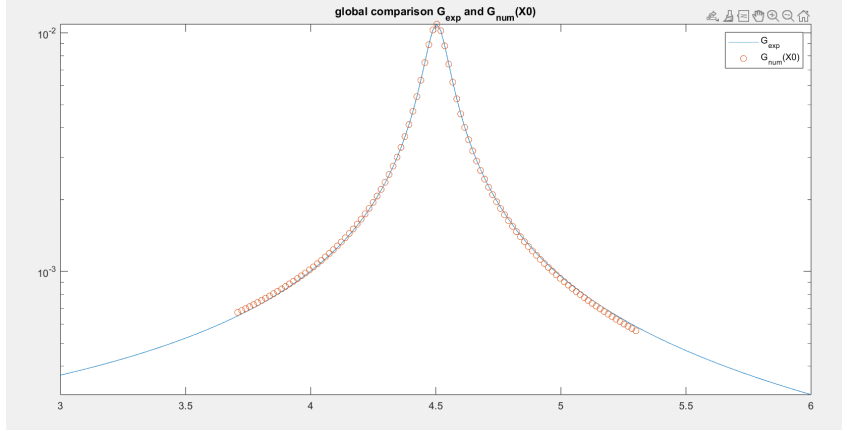
The Experimental Matrix $[G_{exp}]$ is a matrix that collects the Frequency Response Functions given by the sensors applied to the system during a test. In the case studied here, 2 accelerometers are supposed to be applied to the beam ($N = 2$), and so the matrix $[G_{exp}]$ has 2 columns. The values stored in the matrix are complex (magnitude and phase).

4.2. Definition of the Initial Guesses

In order to proceed with the modal identification analysis we have to provide to the minimization algorithm a vector, $\underline{X}_0$ of initial guesses. The vector has a length of $3N + 2$.

- **Guess for ω_i :** obtained from the simplified procedure, with function *findpeaks*;
- **Guess for ξ_i :** obtained from the simplified procedure, looking at the slope of the phase in correspondence to ω_i : $\xi_i = -1/\omega_i \cdot (1/d(phase))$.
- **Guess for A_{jk} :** obtained from the simplified procedure as: $A_{jk} = -imag[G_{exp}(\omega_i) \cdot (2\omega_i^2 \xi_i)]$;
- **Guess for the residuals:** put to zero both for R^L and R^H .

As a first attempt, it is possible to evaluate the accuracy of the guesses by building a numerical model based on $\underline{X}_0$. The result obtained, shown in Figure 4.1 for the first peak of the first FRF, is already quite a good approximation of $[G_{exp}]$ around the resonances.

Figure 4.1: Guess $\underline{X}_0$ for ω_1

4.3. Error function and minimization

In order to define the estimated numerical curve $[G_{num}]$ that fits the experimental data stored in $[G_{exp}]$ we rely on a least square minimization method, using the built-in Matlab function *lsqnonlin*. This function provide us the numerically computed values of ω_i , ξ_i , A_{jk} , and residuals that best fit the data around each resonance peak. In the table below we can compare this outputs with the exact values, that we know from the previous analysis of the beam's FRF.

	EXACT DATA	ESTIMATED DATA
ω_1	28.2983 Hz	28.300 Hz
ω_2	177.3424	177.324
ω_3	496.5637	496.520
ω_4	973.0667	972.840
ξ_1	0.01	0.0100
ξ_2	0.01	0.0100
ξ_3	0.01	0.0100
ξ_4	0.01	0.0100

Table 4.1: Computed/exact comparison

4.4. Graphical outputs obtained

4.4.1. First Accelerometer: results

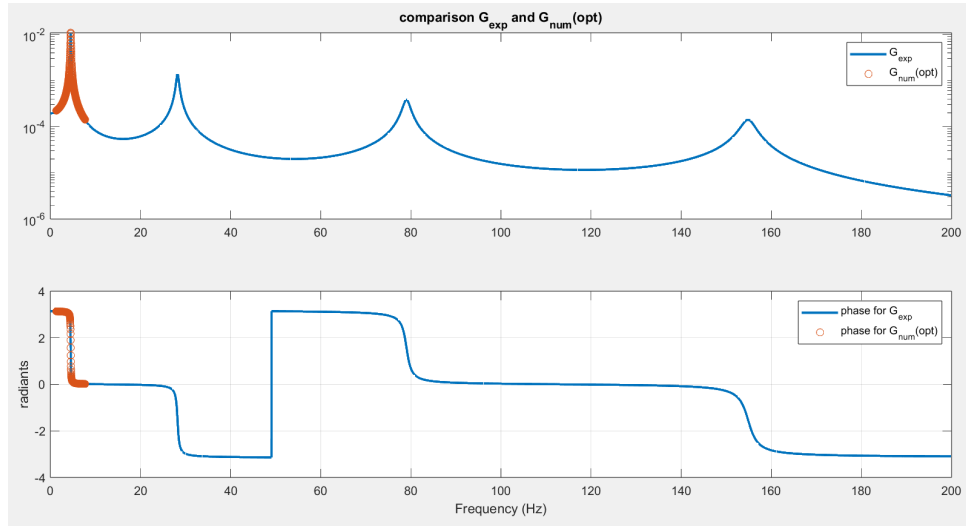


Figure 4.2: First peak, 1 sensor

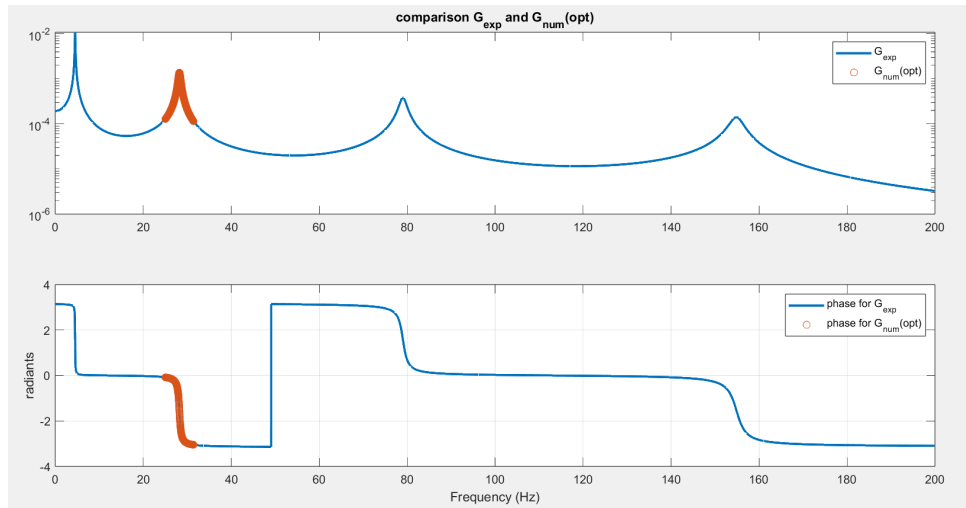


Figure 4.3: Second peak, 1 sensor

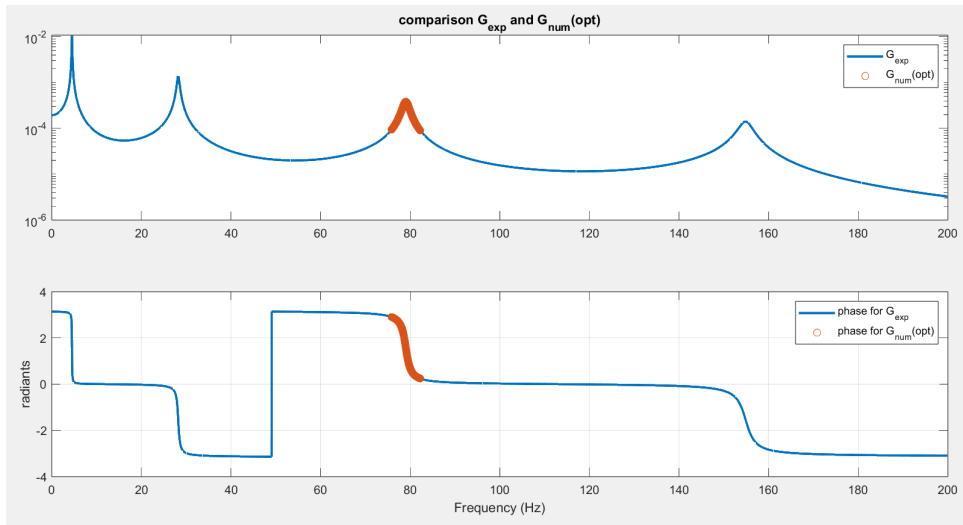


Figure 4.4: Third peak, 1 sensor

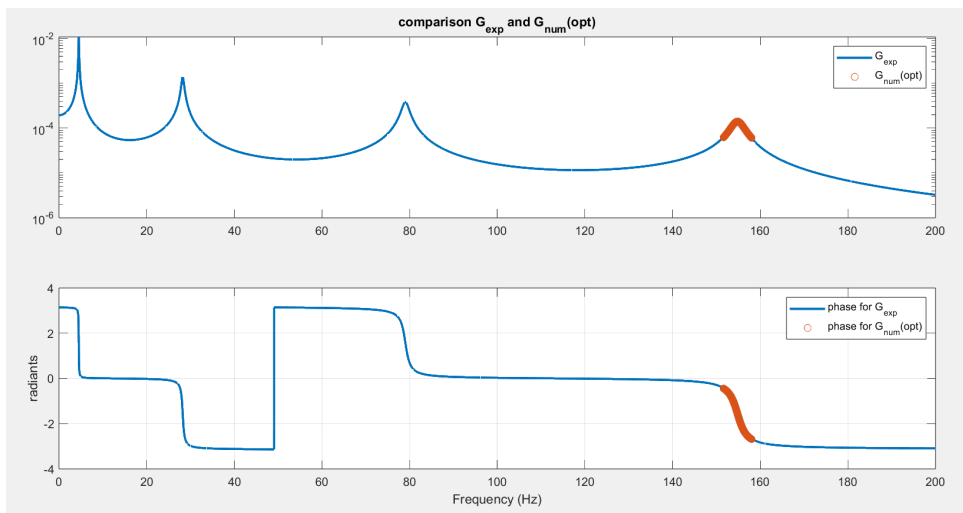


Figure 4.5: Fourth peak, 1 sensor

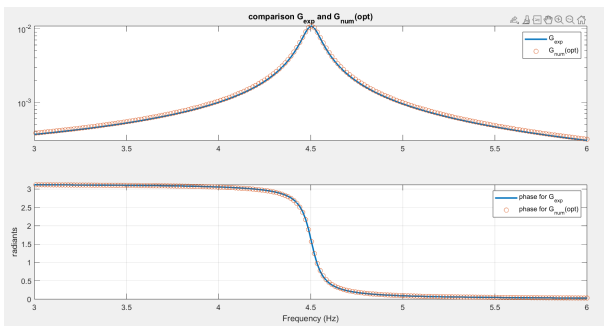


Figure 4.6: First peak: 1 sensor

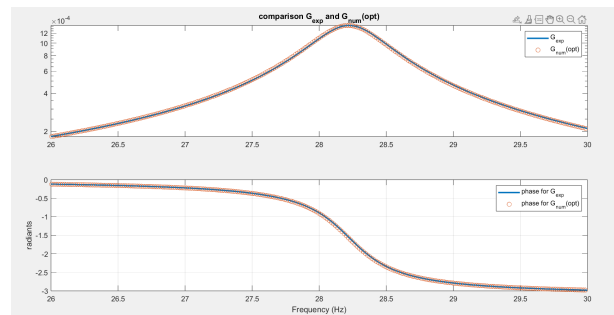


Figure 4.7: Second peak: 1 sensor

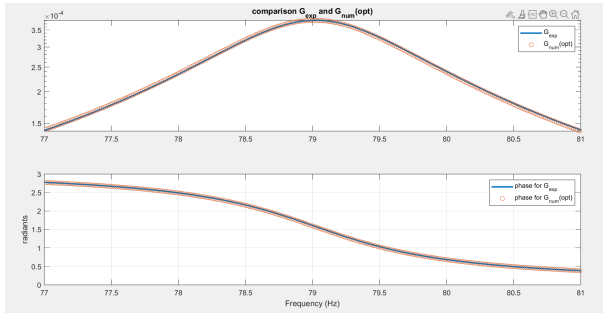


Figure 4.8: Third peak: 1 sensor

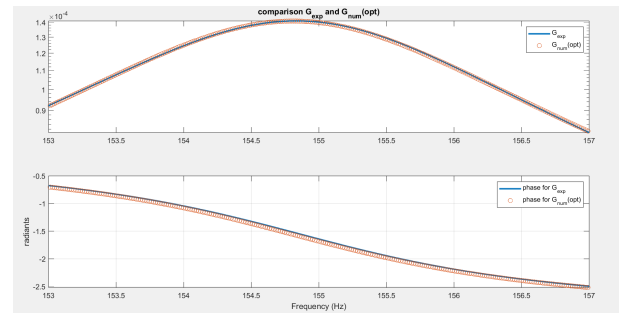


Figure 4.9: Fourth peak: 1 sensor

4.4.2. Second Accelerometer: results

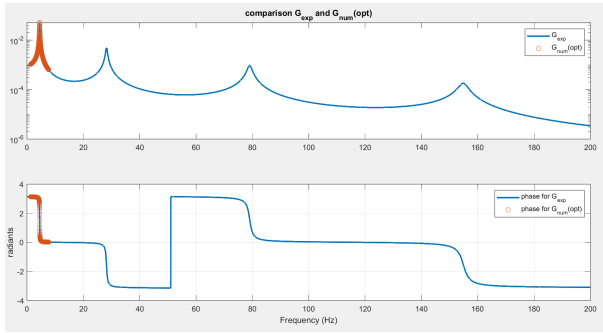


Figure 4.10: First peak: 2 sensor

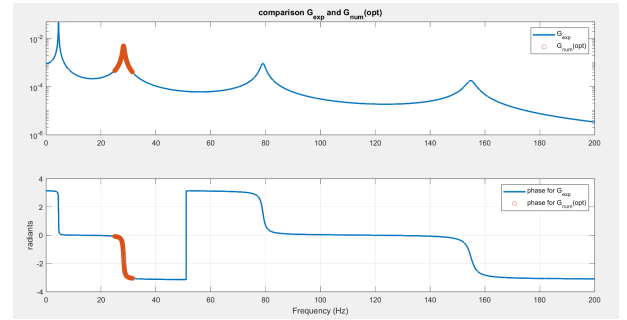


Figure 4.11: Second peak: 2 sensor

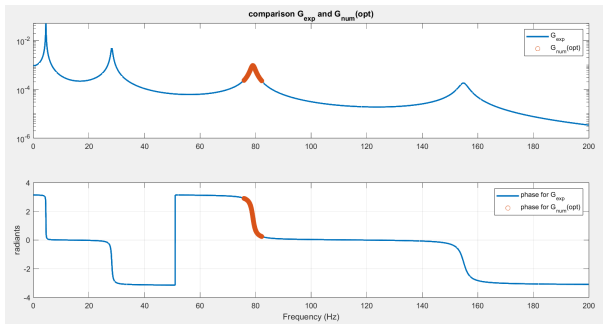


Figure 4.12: Third peak: 2 sensor

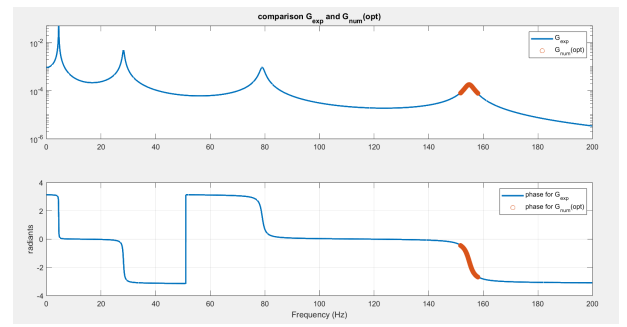


Figure 4.13: Fourth peak: 2 sensor

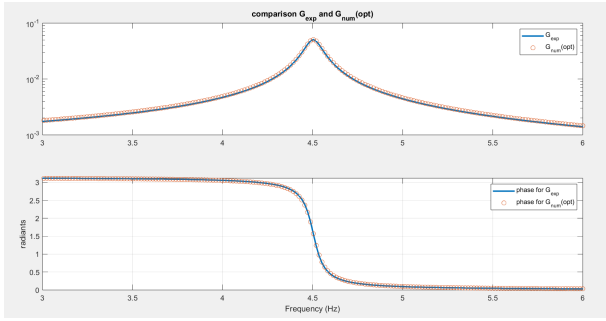


Figure 4.14: First peak: 2 sensor

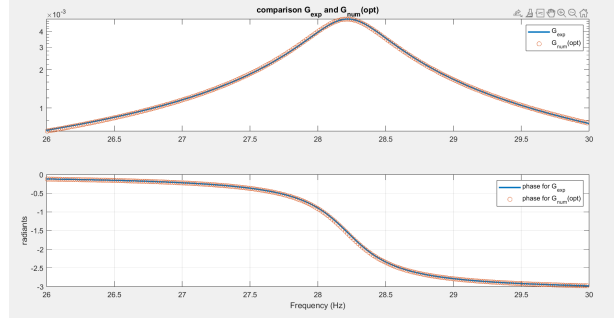


Figure 4.15: Second peak: 2 sensor

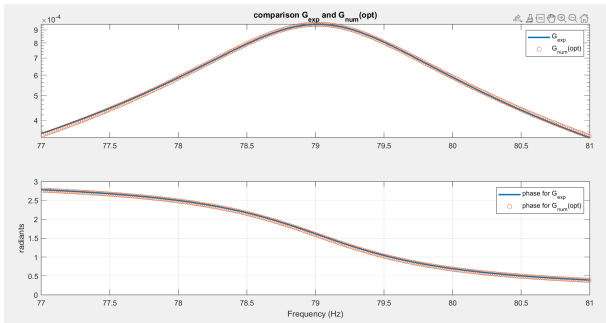


Figure 4.16: Third peak: 2 sensor

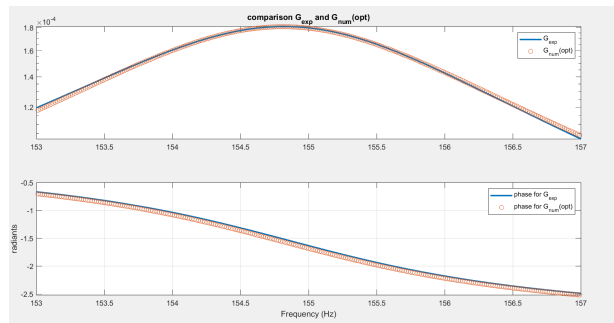


Figure 4.17: Fourth peak: 2 sensor

4.5. Mode Shapes Identified

It's possible to plot the results of the minimization process on the deformed mode shapes presented in Paragraph 2.5, with a proper scaling factor. In that way it's possible to verify the accuracy of the numerically estimated data. Since the $[G_{exp}]$ matrix was generated starting from analytical expressions, we expect that the results will perfectly overlap on the theoretical deformed shapes.

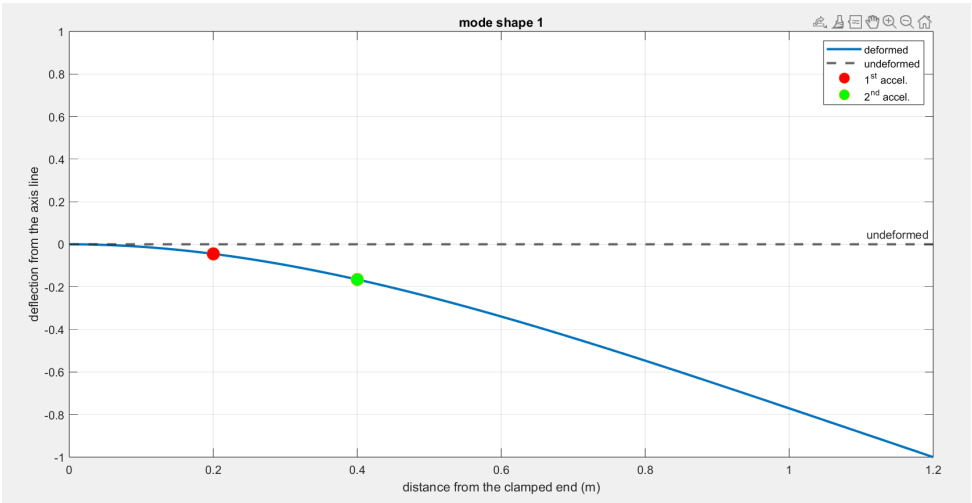


Figure 4.18: First Mode Shape

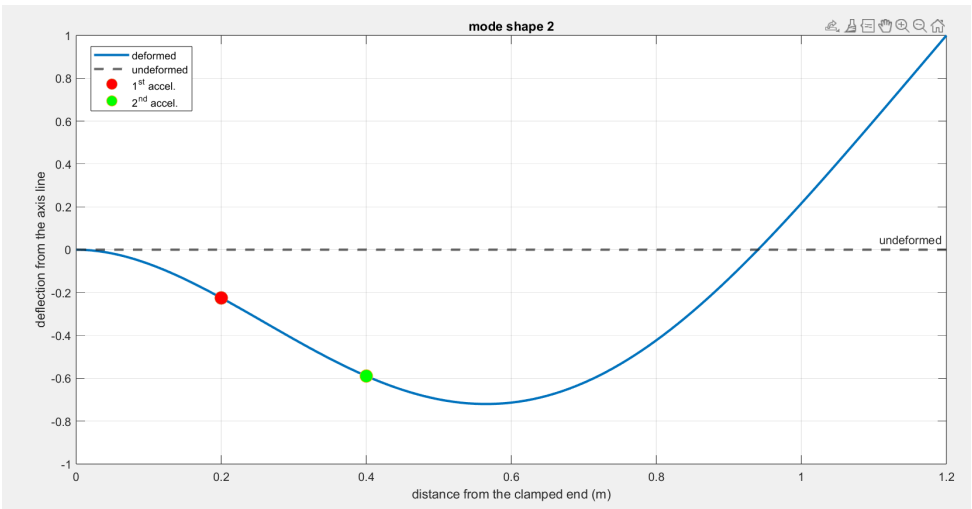


Figure 4.19: Second Mode Shape

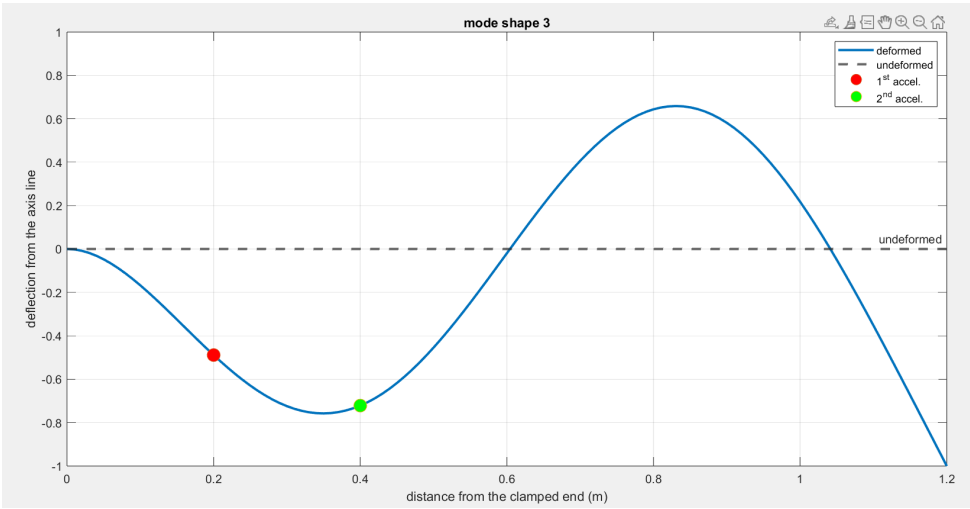


Figure 4.20: Third Mode Shape

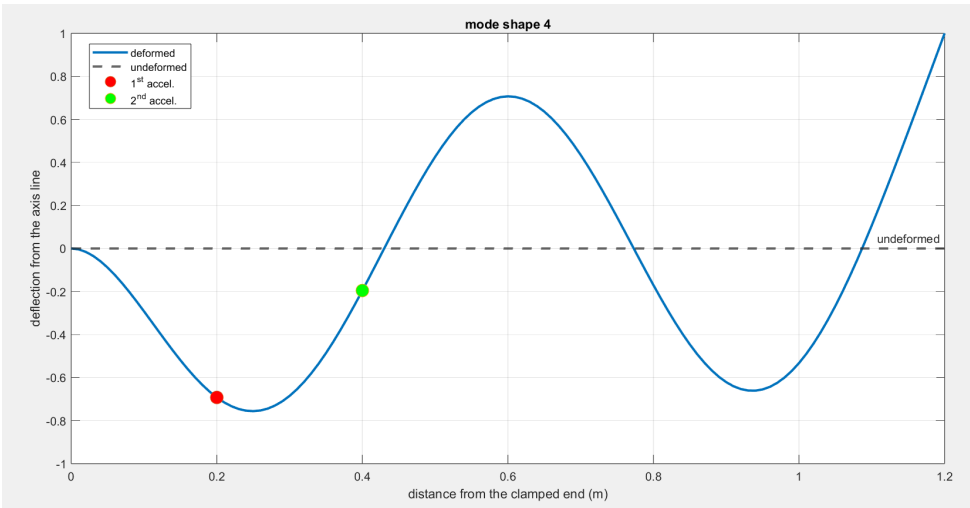


Figure 4.21: Fourth Mode Shape